

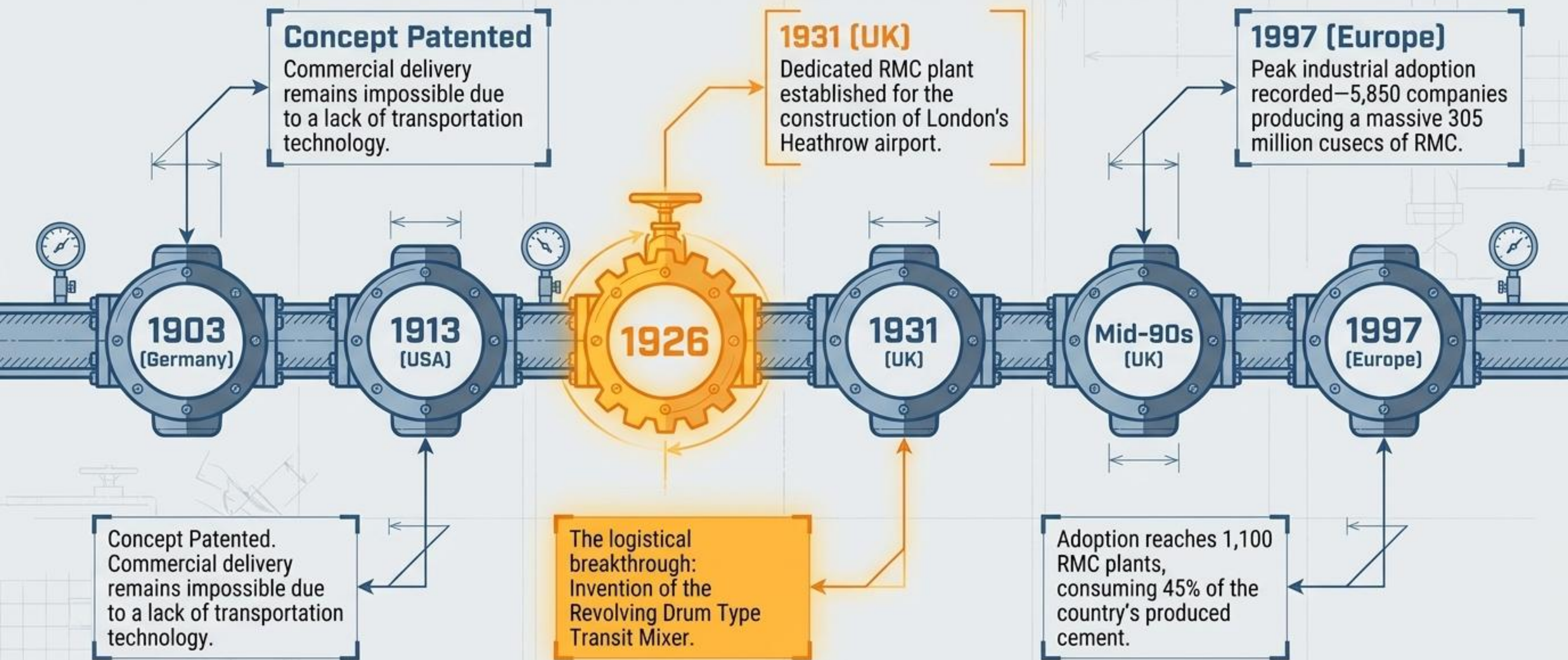


Ready Mix Concrete (RMC)

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THE ON-SITE PROBLEM

	Manual gang mixing
	Congested material storage (godowns)
	High cement wastage and dust pollution
	Slow, labor-intensive energy usage

THE RMC SOLUTION

	Computerized, precise control of aggregates and water per mix designs
	Zero site allocation needed for cement, aggregate, and sand
	Minimized waste and a clean, noiseless, pollution-free environment
	Programmed mechanized delivery enabling speedy construction and project time savings

MATERIALS REQUIRED FOR RMC



CEMENT

The Chemical Binder

Dry powder that reacts chemically with water to bind aggregate particles together.



AGGREGATE

The Inert Foundation

Gravel, sand, and stone that form the bulk volume and structural integrity of the concrete.



FLY ASH

The Sustainable Byproduct

Sourced from coal-fired electricity generating power plants, used to enhance mix properties.



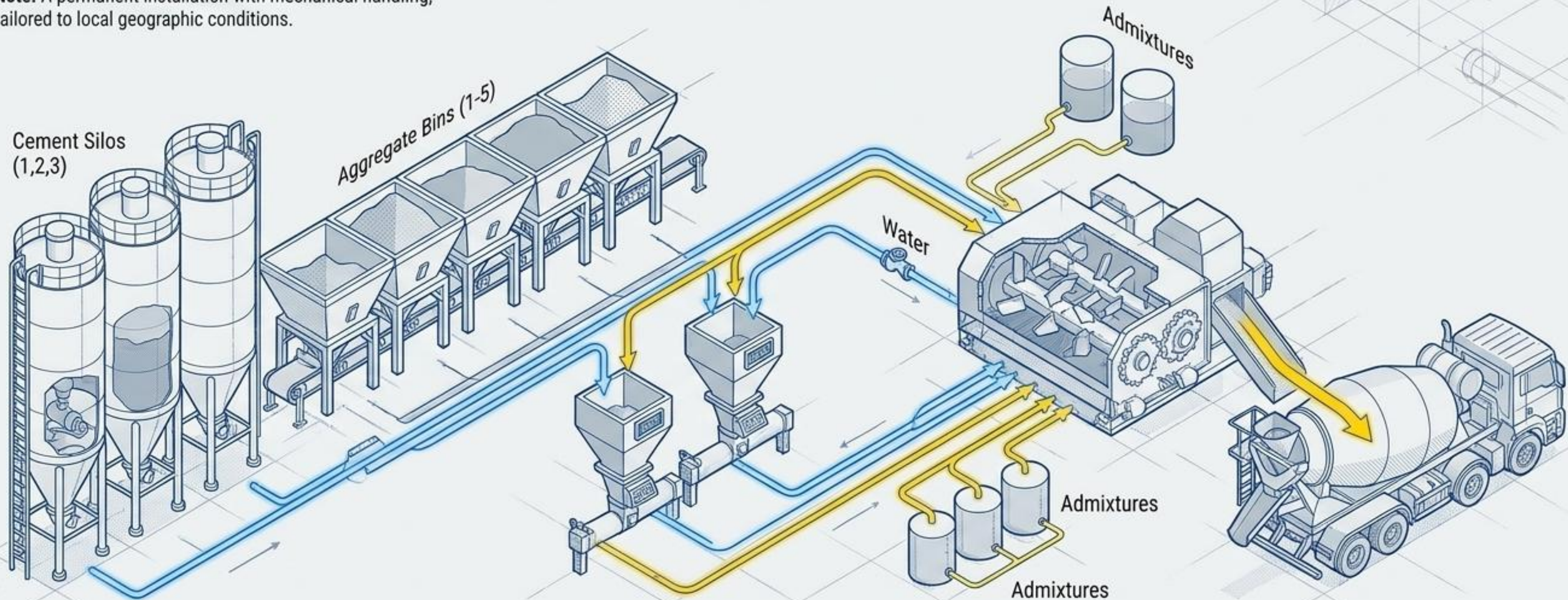
ADMIXTURES

The Chemical Modifiers

Added to alter specific properties (e.g., compensating for high temperatures, loss of air, or long delivery times).

ANATOMY OF AN RMC PLANT

Note: A permanent installation with mechanical handling, tailored to local geographic conditions.



1 STORAGE (INPUTS)

Cement Silos (1,2,3)
Aggregate Bins (1-5)

2 PROPORTIONING

Separate Weigh Hoppers for
Aggregate and Cement to ensure
exact computerized mix designs.

3 PROCESSING

Introduced into the central
Twin-Shaft Mixer.

4 OUTPUT

Discharged directly into the
awaiting transit mixer truck.

THE 3 METHODS OF PRODUCTION

TRANSIT MIXING	CENTRAL MIXING	SHRINK MIXING
WHERE MIXING OCCURS Completely En Route.	WHERE MIXING OCCURS 100% at the Plant.	WHERE MIXING OCCURS Shared (Plant + Truck).
VEHICLE TYPE Truck Mixer.	VEHICLE TYPE Agitator Truck (slow revolution).	VEHICLE TYPE Truck Mixer.
KEY OPERATIONAL BENEFIT Maximum flexibility; water added and mixed on the way to the job site.	KEY OPERATIONAL BENEFIT Prevents segregation; allows more concrete transported per trip, reducing transport charges.	KEY OPERATIONAL BENEFIT Plant initiates partial mix; truck finishes the job to ensure uniformity.

TRANSIT MIXING LOGISTICS: THE ROUTE

AT PLANT

Dry materials are proportioned and loaded into a mixer drum mounted on a conventional chassis.



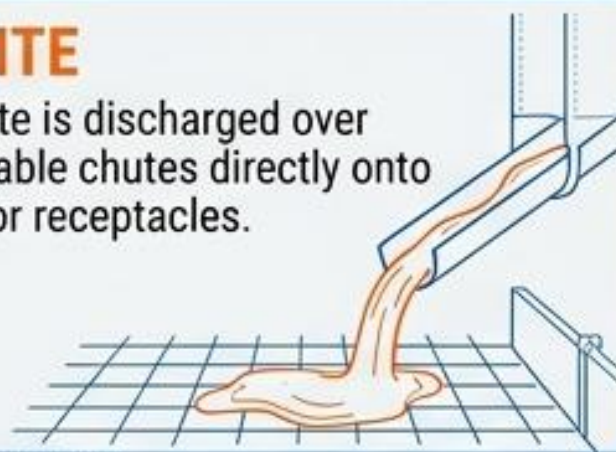
EN ROUTE

The driver takes control. They operate the water level and mixing controls. Power is drawn from the truck engine or an independent petrol engine.



AT SITE

Concrete is discharged over detachable chutes directly onto forms or receptacles.



TIMING OPTIMIZATION

For long hauls, mixing is purposefully delayed and started towards the very end of the haul to assure delivery of freshly mixed concrete.



Point A
(The Plant)
51°09'413" N
36°10'533" E



Point B
(The Job Site)
80°04'467" N
33°29'567" E

Central Mixing (The Agitator)



The plant completes proportioning and mixing. The truck acts merely as an agitator, revolving slowly to prevent "segregation" and initial stiffening.

Strict Constraint:
Maximum 1.5 hours allowed between
mixing and placing.

Shrink Mixing (The Finisher)



The stationary plant-mounted mixer partially mixes the concrete. The truck completes the process.

Standard Requirement:
Approximately 30 turns in the truck
drum, or 2 minutes at mixing speed,
to achieve total uniformity.

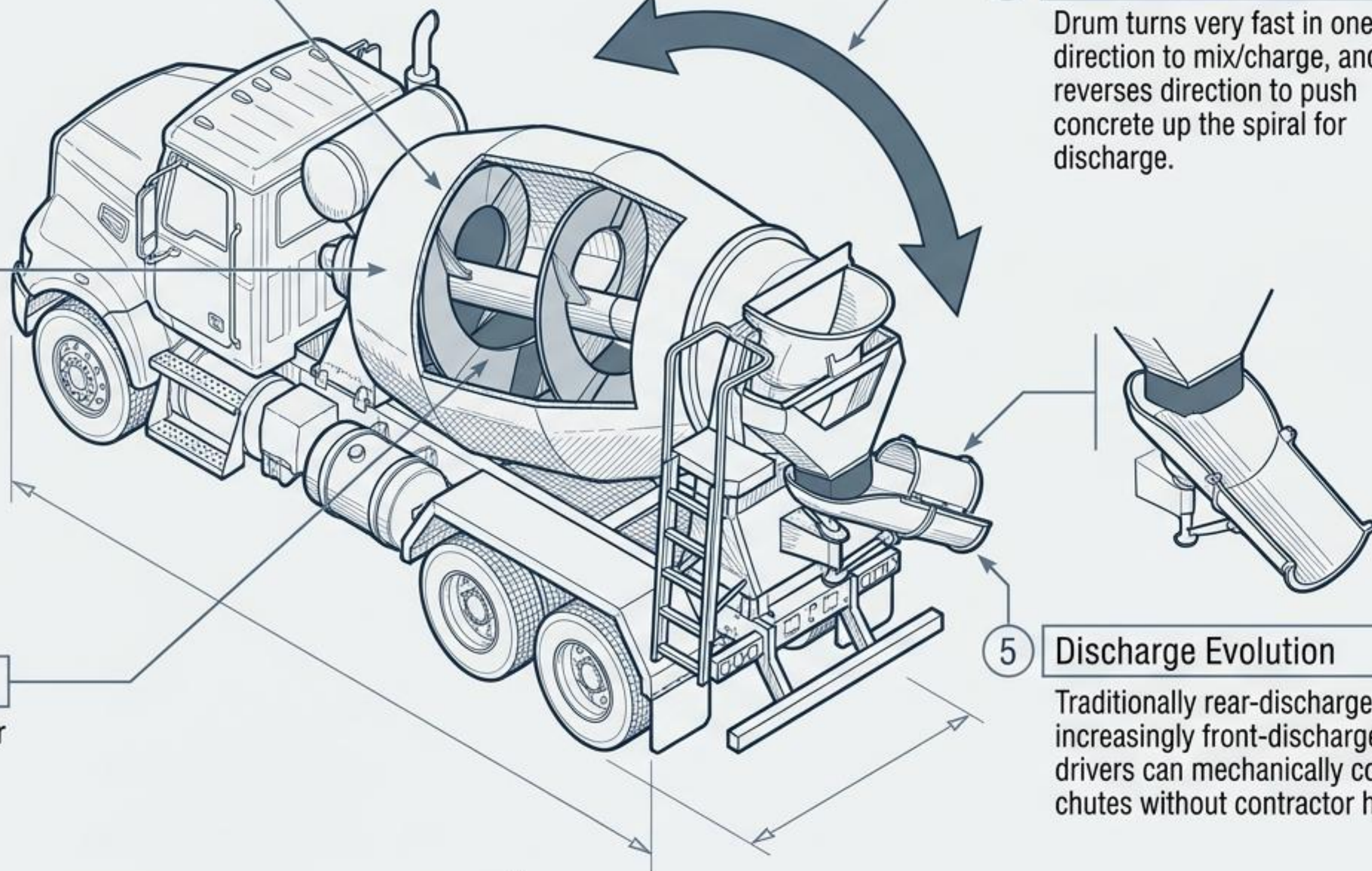
1 **The Blades**
Helical (spiral) configuration wrapping from head to opening.

2 **Capacity Limits**
Typical size is 9 to 11 yd³. Drums are strictly limited to a max 63% gross volume when mixing, and 80% when acting as an agitator.

3 **The Drum Axis**
Inclined to the horizontal for optimal material retention and flow.

4 **Dual-Direction Mechanics**
Drum turns very fast in one direction to mix/charge, and reverses direction to push concrete up the spiral for discharge.

5 **Discharge Evolution**
Traditionally rear-discharge; increasingly front-discharge so drivers can mechanically control chutes without contractor help.



Logistics & Compliance Dashboard

< 2 RPM

The maximum speed the drum turns while hauling to the job site.

90 Minutes

ASTM C 94 Specification: Concrete must be discharged within 90 minutes of water touching the cement.

300 Revolutions

The absolute maximum number of drum revolutions allowed before discharge.

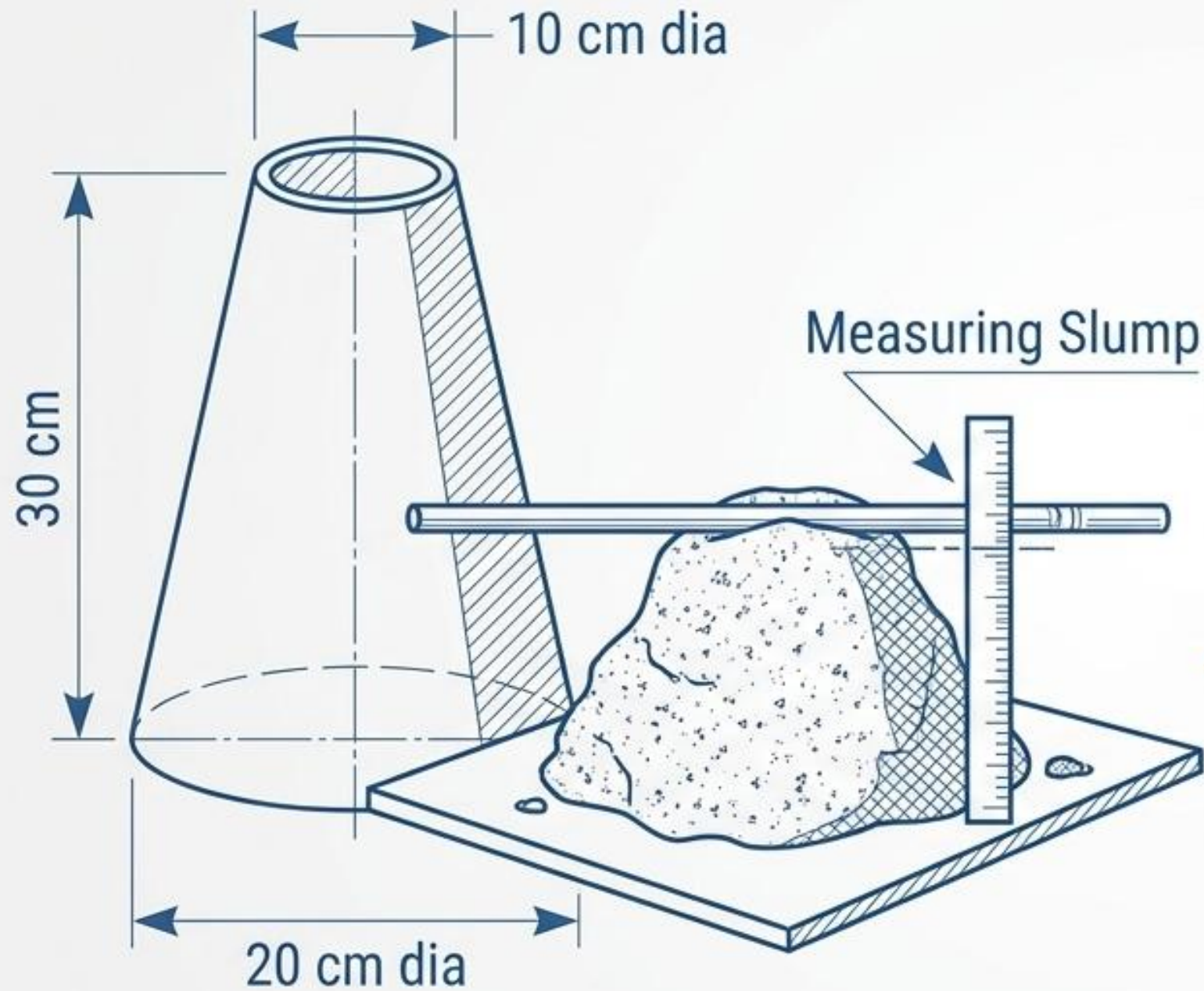
30 Turns / 2 Mins

The minimum required mixing speed turns if water is legally added at the site to adjust mix.

Quality Control at the Site: The Slump Test

The Golden Rule of Adjustments

Water should *never* be added to the mix unless the measured slump is strictly less than specified. If added, it must go in all at once, followed by a minimum of 30 revolutions (or 2 minutes) of mandatory mixing.



Operational Challenges & Limitations



Logistical Friction

Double handling of materials (to the plant, then to the site) results in additional baseline costs and weight losses.

Storage godowns require large physical footprints at the plant.



Environmental Contamination

Wind, weather, and site mishandling can allow impurities to creep into aggregate reserves.



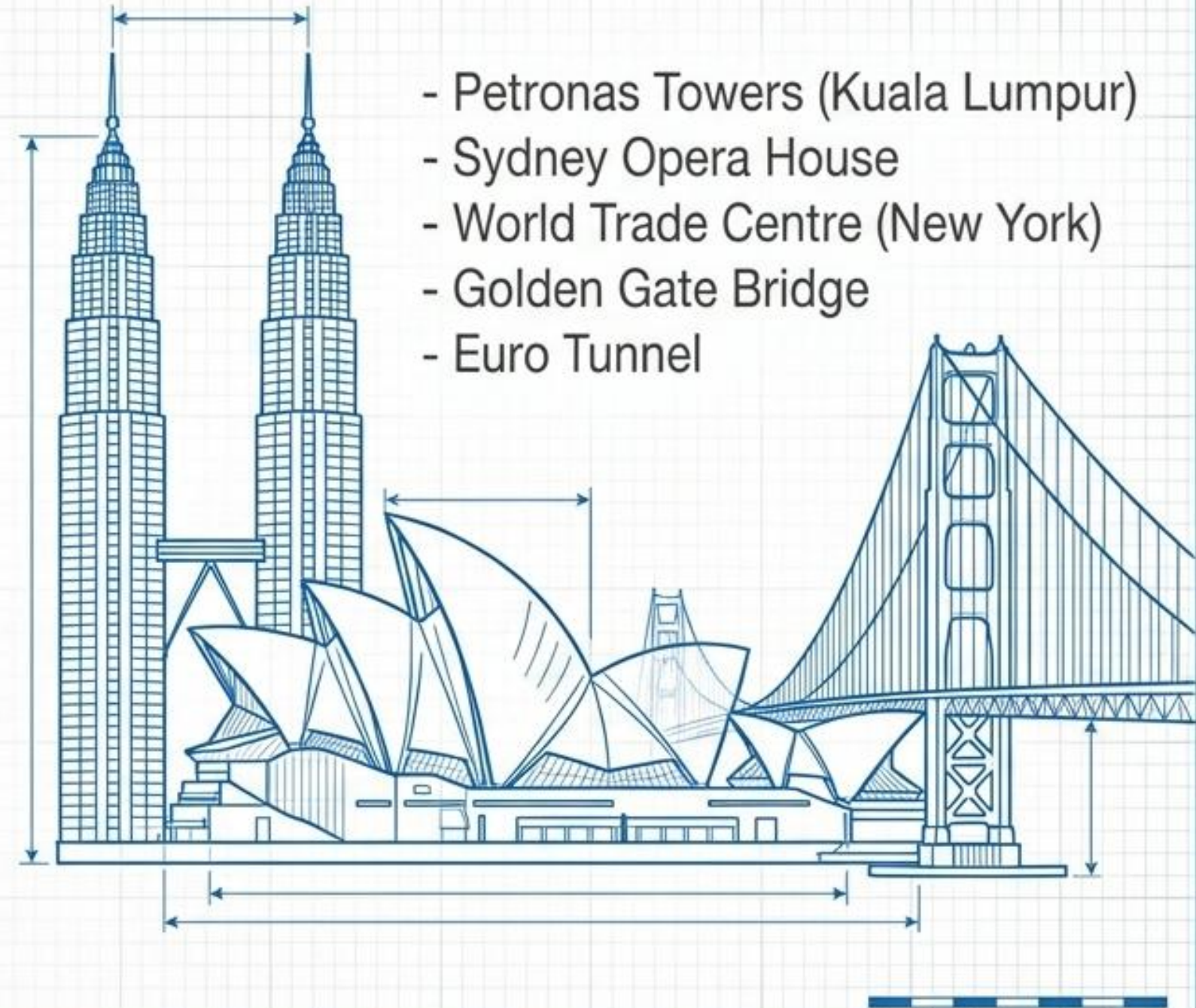
Human Vulnerability

"The Mercy of Gangs." The highly precise plant mix can be ruined at the site by unauthorized manipulation of the water-cement ratio or manual errors during placement.

Built in India

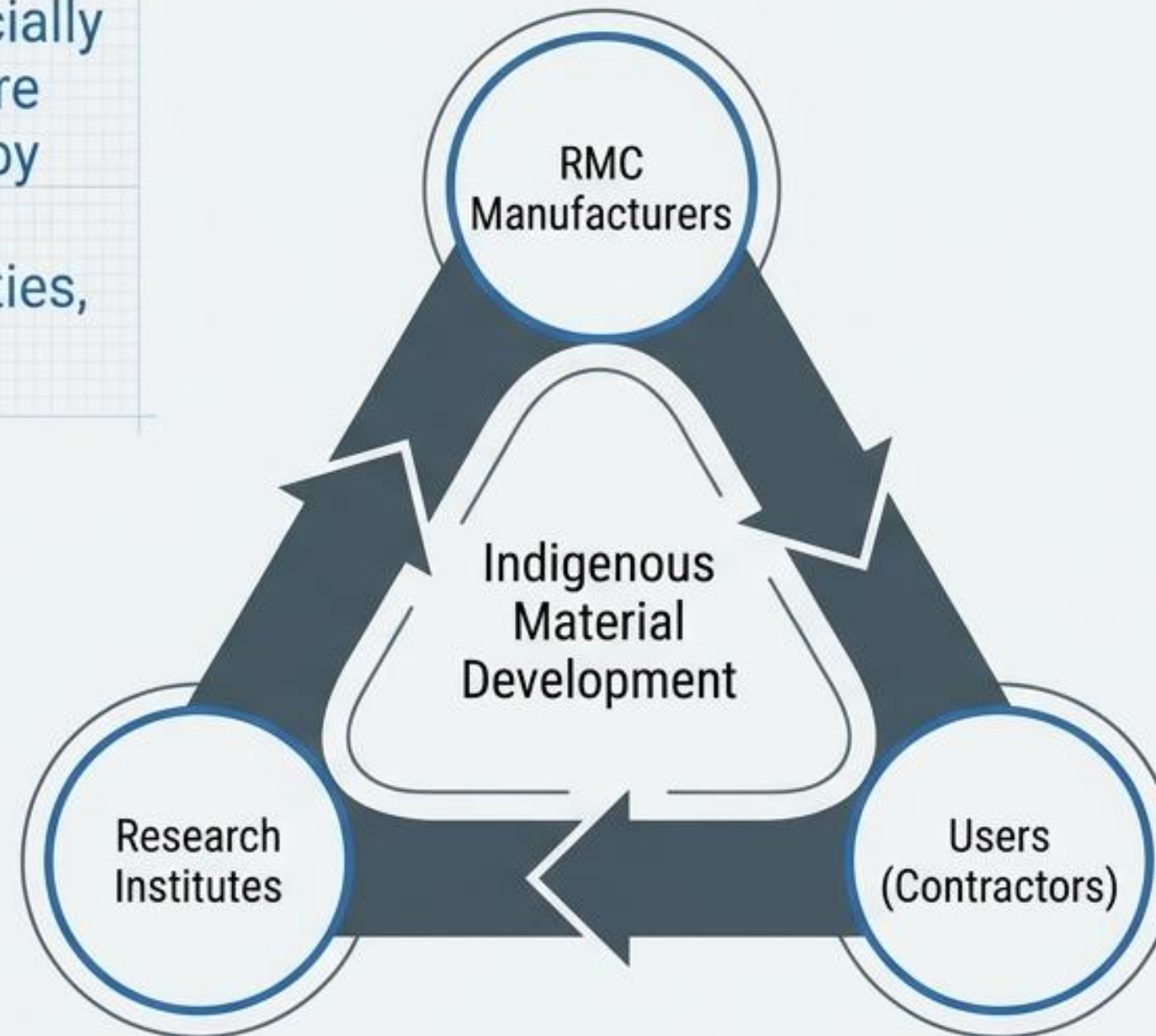
- The Mumbai-Pune Expressway (95kms)
- 55+ Mumbai flyovers
- Chennai Metro station superstructures
- Tidal Park Chennai
- Worli Outfall (M45 grade RMC with 30% OPC)

Global Icons



The Future Ecosystem of RMC

The future is bright, especially in developing infrastructure markets like India (aided by government concessions on land, utilities, and taxes).



The 3 Pillars of Success

1

Regular, uninterrupted supply of raw materials.

2

Highly efficient operation of permanent plant machinery.

3

Uncompromising, guaranteed production of concrete strength and quality.